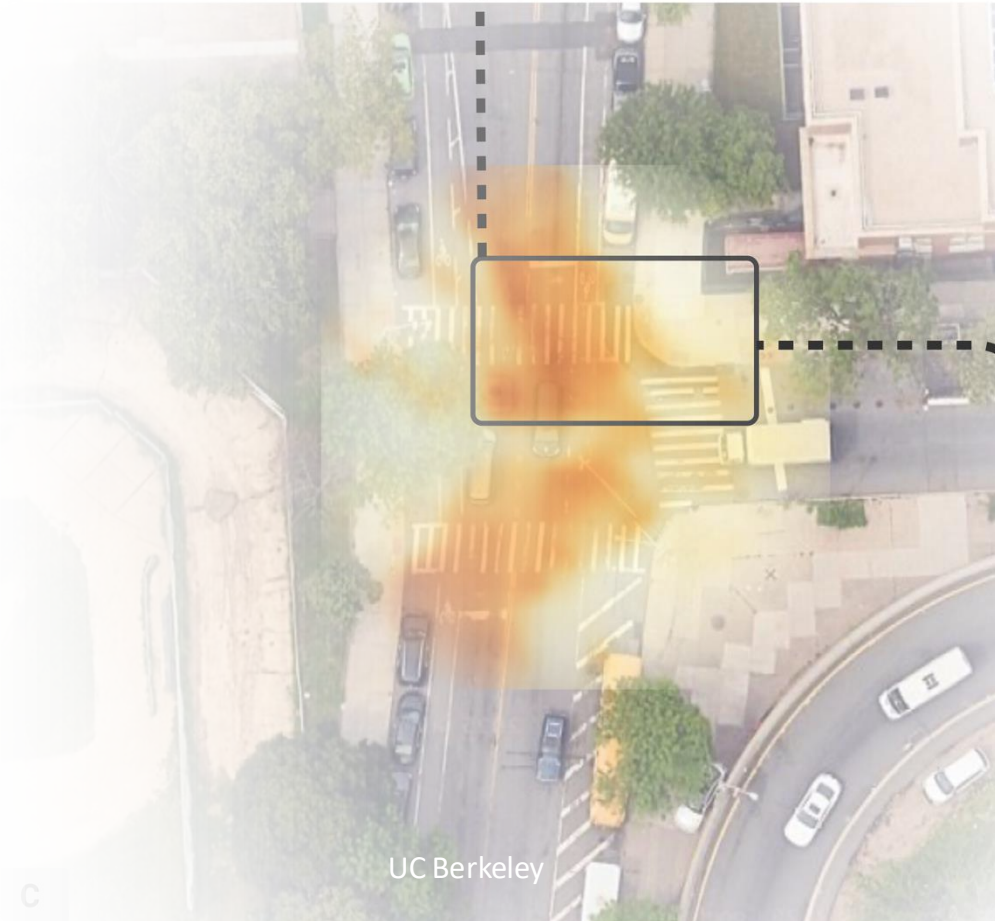
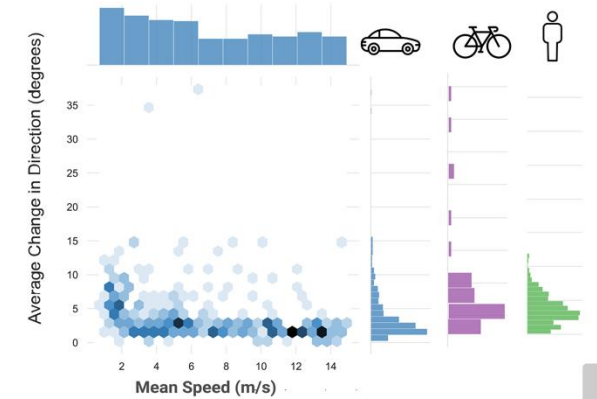
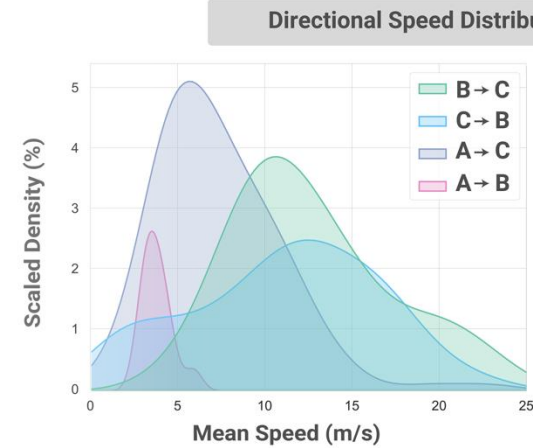
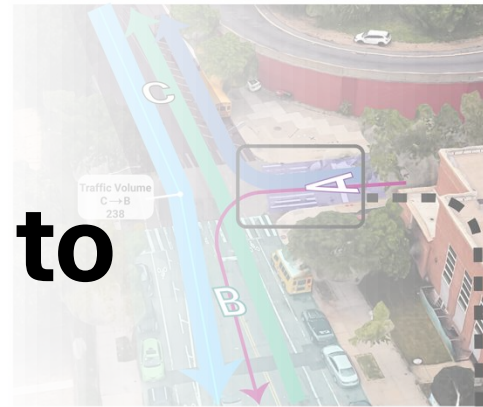


Introduction to Urban Data Analytics

Fall 2026

Maryam Hosseini

City & Regional Planning



UC Berkeley

C

B

D

Course Overview

Topics, Survey results, Workload,

This Course Does Not Cover

- ArcGISPro / QGIS Software
- Statistical modeling, hypothesis testing
- Time series analysis
- Machine Learning
- Other Programming Languages (R, Java Script,)

This Course Does Cover

- Intro to urban Systems as a complex system
- Intro to GIS
- Intro to Computational Thinking
- Geospatial analysis in Python
- Data Visualization fundamentals
- Geospatial Data Visualization
- Deceptive Visualization & Misinformation

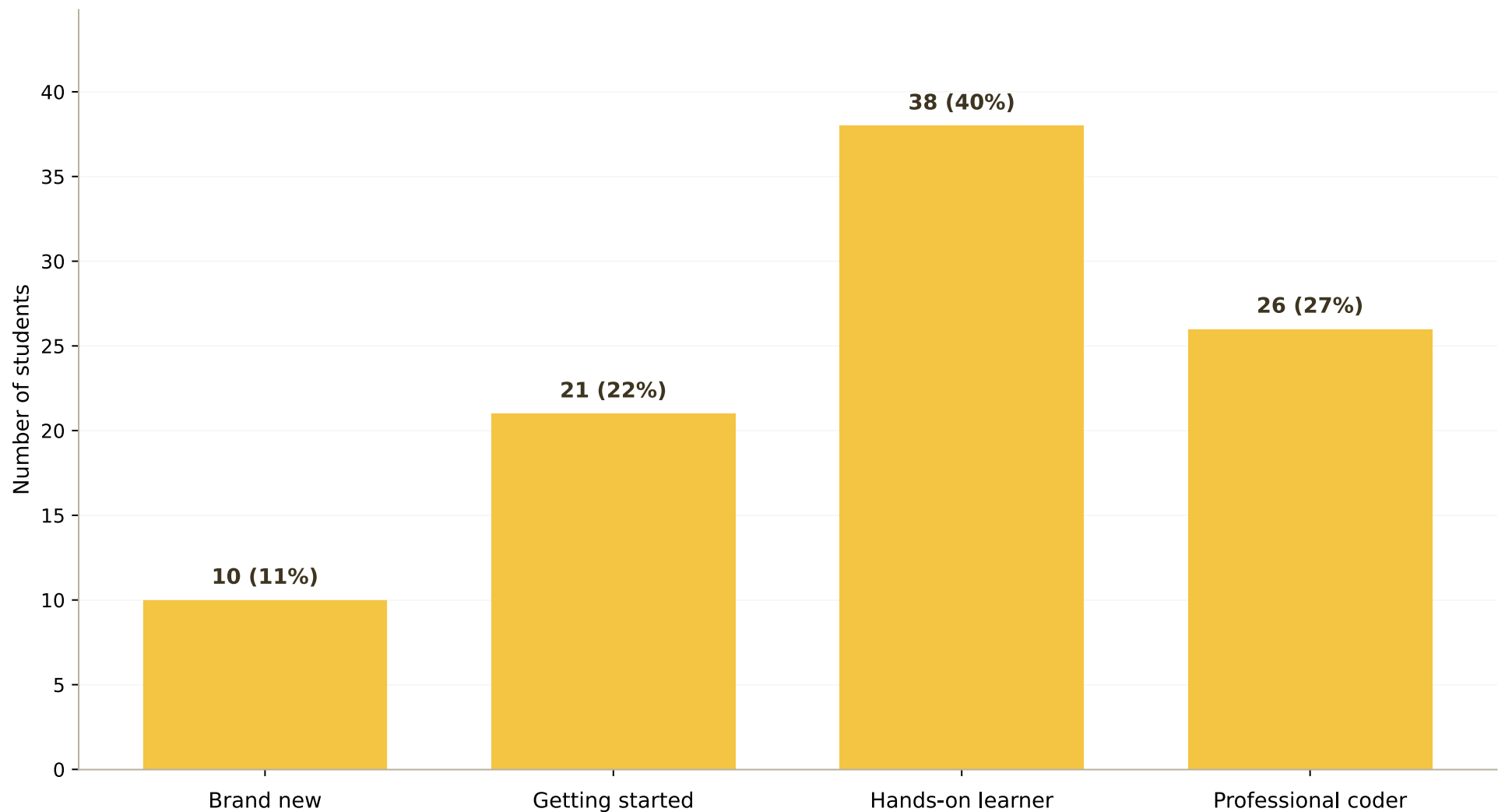
Checkout course schedule on the [course website](#)

This **is not** an intro to programming course!

- Python basics ($\approx$ 2 weeks)
- Transition to pandas and geopandas
- Principles of effective data visualization
- Creating expressive visualizations and maps in Python
- Exploring multiple libraries for geospatial visualization
- Hands-on analytics with real-world data
- Data sanity checks, cleaning, and implications
- Geospatial operations in Python

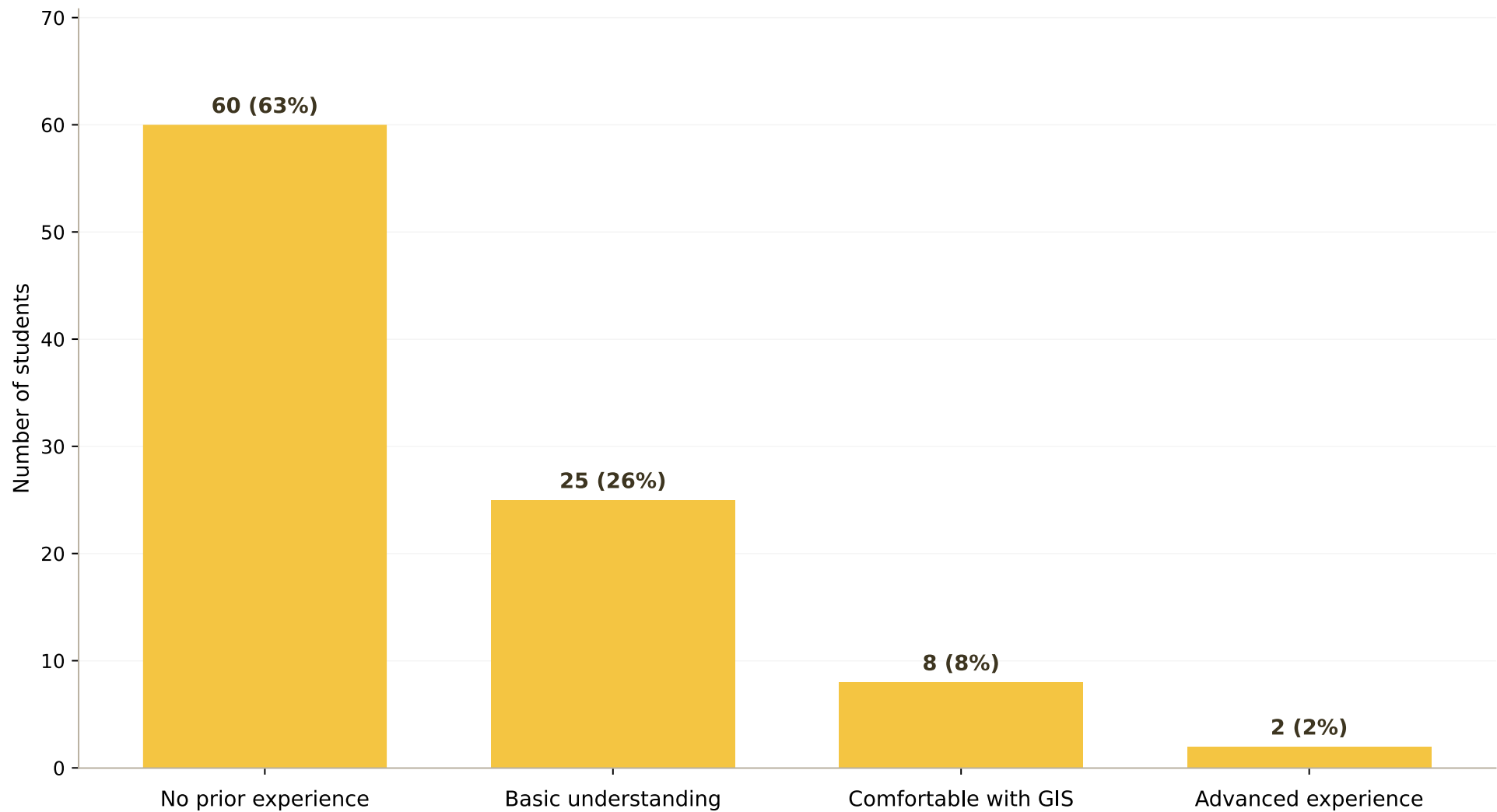
Survey Results

How would you describe your coding experience?



Survey Results

How would you describe your experience with GIS and geospatial data?



Survey Results

Expectations



Weekly Time Commitment: **7.5 Hours** Outside Work

- **This course requires approximately **7.5 hours** of work outside of class each week.**
- Those who answered less than 3 hours: I recommend reconsidering your enrolment

Course Logistics

Attendance, Assignments, Final Project

Components

- Lectures
- Class Activities → Completion
- Lab Notebooks → Completion
- 4 Assignments (Individual) → Autograded for correctness
- No Exams
- Final Project (Groups)
 - Proposal + Lightning Talk Video
 - Midterm Submission
 - Final Submission

Opportunity to work with a real client on a real-world problem

Infrastructure

- **Resources**

Course website

- **Coding**

JupyterHub on DataHub

- **Submission**

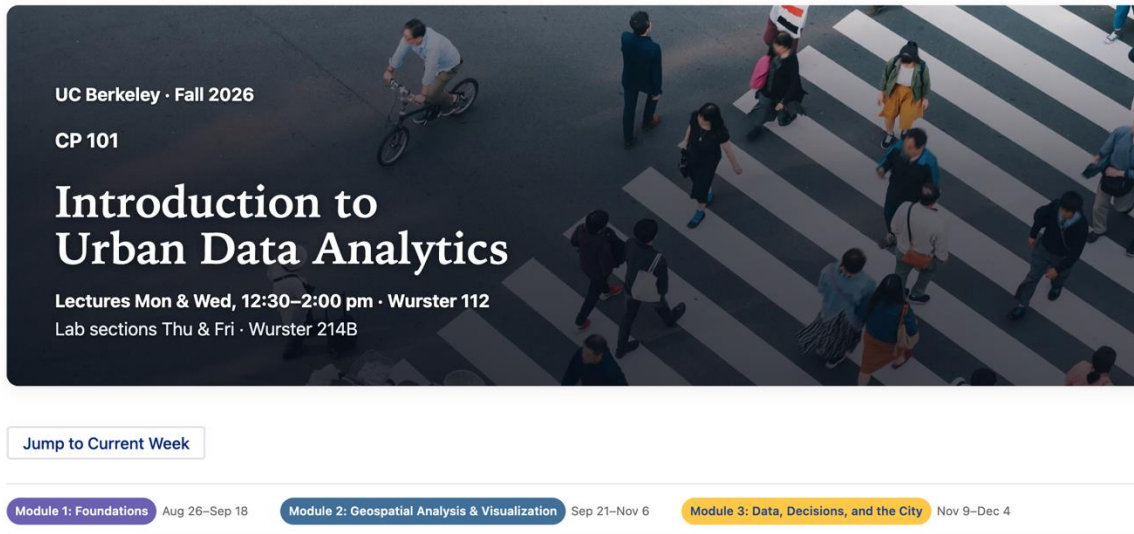
Gradescope

- **Questions, Discussion & GSI Support**

Ed Discussion

Course Website

<https://ucb-cp101.github.io/fa26/>



Main source of information about this course. The site will be updated frequently.

- **Bookmark It:** Add the website to your bookmarks and check it regularly for announcements and materials.
- **Readings and Resources:** Lecture slides, reading and videos and extra materials
- **Assignments, Labs and Class Activities**
- **Syllabus** and Course Policies and

Attendance Policy

- Your presence matters. Your active participation matters even more!
- You may take up to **4 absences (class & lab)**, no questions asked.
- If you will miss a class or lab, please **email all three members of the teaching team** in advance.
- If an emergency requires additional absences beyond the four, email us **as soon as possible** and include supporting documentation.

- **Lecture participation and in-class activities: 15%** of your final grade.
- **Lab participation and activities: 15%** of your final grade.

Grading Breakdown

- **Class attendance and activities (15%)**
- **Lab attendance and activities (15%)**
- **Assignments (4 Graded): 30%**
 - Dropping the lowest grade for submitted work
 - Unsubmitted work/empty submission receives zero and cannot be dropped.
- **Final Project: 40%**
 - Proposal : 5%
 - Midterm Submission: 15%
 - Final Submission: 20%

Late Submission Policy

Extensions

- 2 extension tokens (24hrs each) for **assignments**
- 1 extension token (72hrs) **final project**

Late Submissions (Without Extensions)

- **Penalty: 10%** deduction **per day** will be applied to late submissions without an approved extension.

Final Project

Track A: Python for Data Interpretation & Visual Reasoning

For students interested in interpreting urban data through maps and visual analysis. Lighter coding focus.

Deliverables

- ArcGIS StoryMap (or equivalent)
- 4+ standalone interactive HTML maps generated with Python
- A PDF report

Emphasis

- Visual reasoning and interpretability
- Thoughtful use of spatial methods, robust cleaning process
- Problem definition, story telling and visualization design judgment

Track B: Python for Data Analytics & Reproducibility

For students more comfortable coding in Python and aiming to showcase technical skills. Shorter written reports in interim phases.

Deliverables

- GitHub repository + GitHub Pages webpage
- 2+ standalone interactive and animated HTML maps generated with Python (integrated in the webpage)

Emphasis

- Reproducible, modular workflows
- Handling large or more complex datasets
- Robust analytical methods

Opportunity Strong projects may be developed into workshop submissions, conference posters, or journal publications.

Final Project

Where can I read about Final Project?

On the website, click **Final Project** in the **top navigation bar**.

Syllabus

Due Dates

Final Project

Office Hours

Final Project Due Dates

Group Formation

Sep 9 – 12:00 pm

Proposal and Lightning Talk Video

Sep 21 – 12:00 pm

Midterm Submission

Nov 2 – 11:59 pm

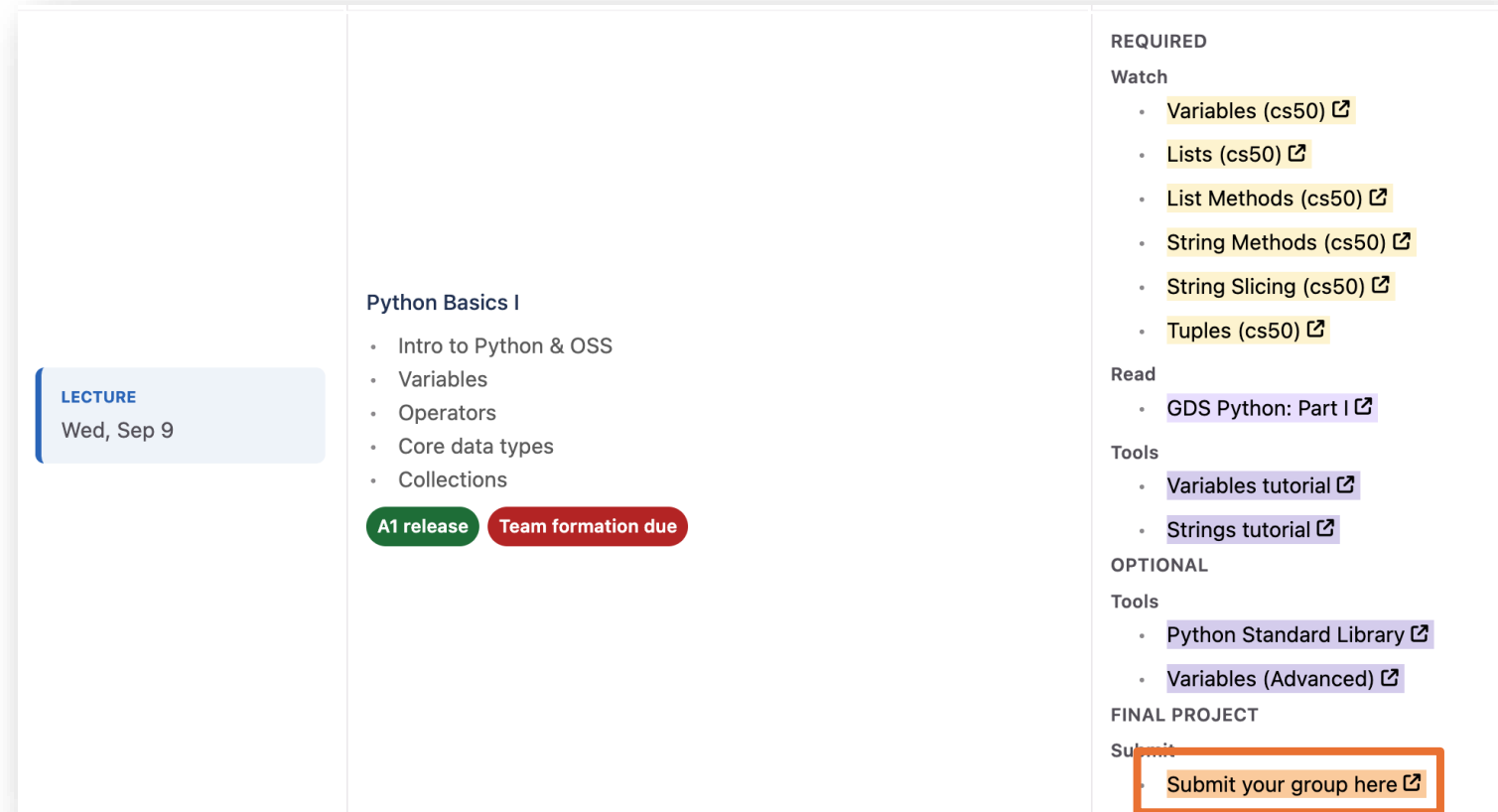
Final Project Submission

Dec 15 – 11:59 pm

Final Project

Group Formation

Due Sep 9 – 12:00 pm



The screenshot shows a course page with three main columns. The left column has a blue box labeled 'LECTURE' with the date 'Wed, Sep 9'. The middle column is titled 'Python Basics I' and lists topics: 'Intro to Python & OSS', 'Variables', 'Operators', 'Core data types', and 'Collections'. Below this list are two buttons: a green one labeled 'A1 release' and a red one labeled 'Team formation due'. The right column is divided into sections: 'REQUIRED' with a 'Watch' subsection listing links for 'Variables (cs50)', 'Lists (cs50)', 'List Methods (cs50)', 'String Methods (cs50)', 'String Slicing (cs50)', and 'Tuples (cs50)'; a 'Read' subsection with a link for 'GDS Python: Part I'; a 'Tools' subsection with links for 'Variables tutorial' and 'Strings tutorial'; an 'OPTIONAL' section with a 'Tools' subsection listing links for 'Python Standard Library' and 'Variables (Advanced)'; and a 'FINAL PROJECT' section with a 'Submit' subsection. The link 'Submit your group here' is highlighted with an orange box, and an orange arrow points from the text 'Where to submit?' below to this link.

LECTURE
Wed, Sep 9

Python Basics I

- Intro to Python & OSS
- Variables
- Operators
- Core data types
- Collections

A1 release Team formation due

REQUIRED

Watch

- Variables (cs50)
- Lists (cs50)
- List Methods (cs50)
- String Methods (cs50)
- String Slicing (cs50)
- Tuples (cs50)

Read

- GDS Python: Part I

Tools

- Variables tutorial
- Strings tutorial

OPTIONAL

Tools

- Python Standard Library
- Variables (Advanced)

FINAL PROJECT

Submit

- Submit your group here

Where to submit?

Optional: MTC-Mentored Project Track

- **Choose one of the 10 project topics** presented by MTC
MTC slides are available on the course website (week 1)
- **Submit your proposal** to be reviewed by MTC (due Sep 21)
Same proposal structure and rubric
- **MTC will select 6–7 teams** based on project fit and feasibility.
- **Grading stays with the course team:** MTC selects teams but does **not** grade proposals or final projects.
- **No extra credit:** all projects are assessed using the same course criteria.

Learning Tips



-
- Have a study buddy or group
 - Watch the videos together
 - Pause and type the codes!
 - Try to explain each new concept to your peer(s) in your own words.
 - Typing out code and explaining while you code are evidence-supported ways to learn programming more effectively than passive watching alone

Have Questions? Feeling Confused?

- **Ed Discussion:** Check before asking
- **Ed Discussion private message** if it's something you don't feel comfortable sharing publicly.
- Book an **office-hours** slot for anything that's easier to talk through in-person.



Cities are Complex Systems

Aerial view of intersection in Ginza, Tokyo, Japan at night

By [tawatchai1990](#)

Environment

Economic

Housing

Transportation

Infrastructure

Public Health

Social

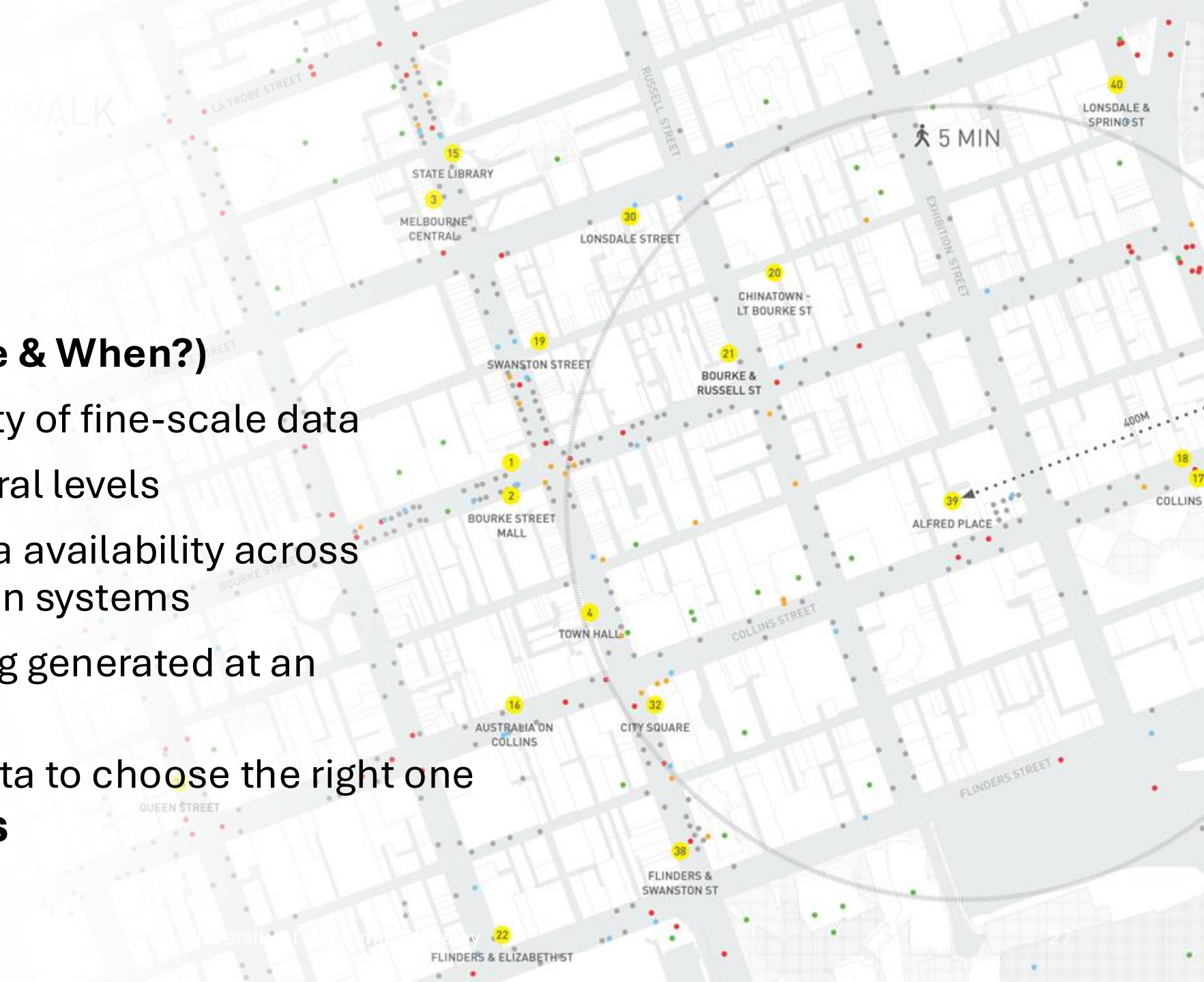


The background of the slide is a dark, textured world map. The map is composed of a dense network of small, light-colored icons and lines. These icons include various symbols such as people, houses, cars, and location pins, which are interconnected by a web of thin lines, suggesting a global network or data flow. The overall aesthetic is high-tech and data-driven.

The Expanding Landscape of Urban Data

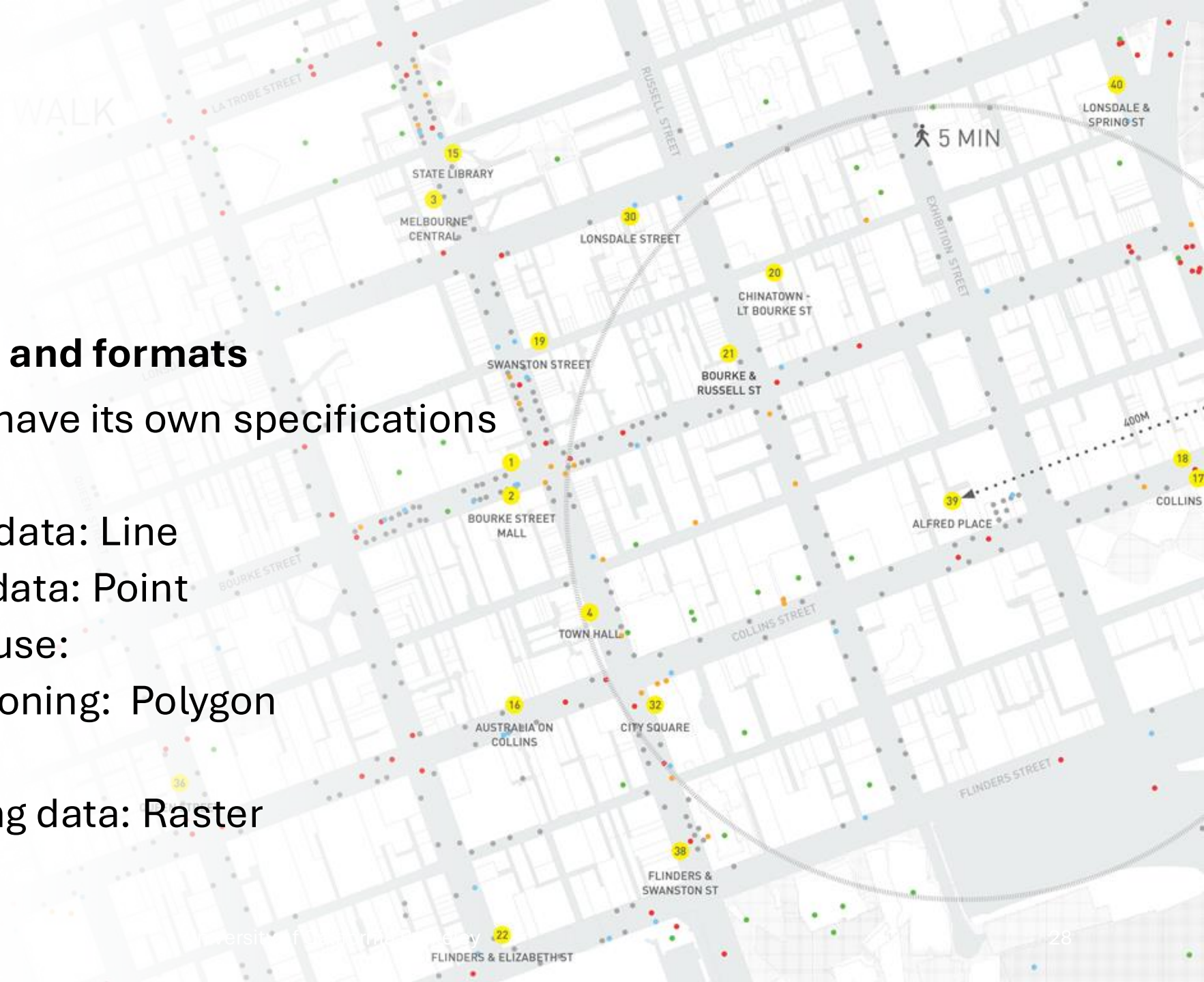
Urban Data

- **Space + Time (Where & When?)**
- Surge in the availability of fine-scale data
- Varying spatio-temporal levels
- Still a large gap in data availability across different parts of urban systems
- Synthetic data is being generated at an unprecedented level
- Understanding the data to choose the right one for our task, **matters**



Urban Data

- **Various data models and formats**
- Each subsystem can have its own specifications
 - Transportation:
 - Road network data: Line
 - Incident level data: Point
 - Housing and landuse:
 - Landuse and zoning: Polygon
 - Environmental:
 - Remote sensing data: Raster





Thinking Systematically, Thinking Computationally

NYC Vision Zero Evaluation

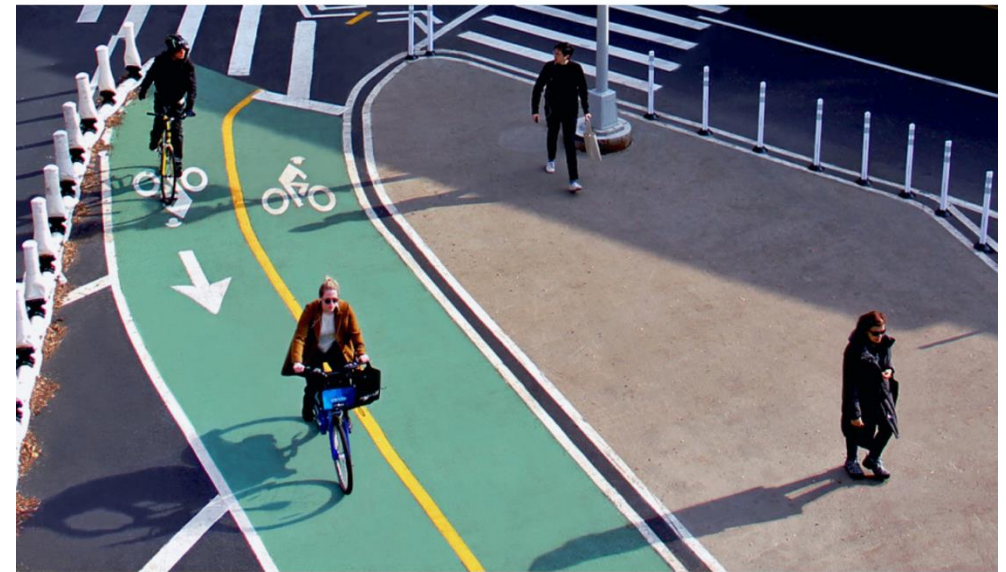
NYC is your client, they want to investigate:

Has Vision Zero improved the situation in NYC since its implementation?



[adopted from Enrico Bertini]

Since 2014, Vision Zero in New York City has brought traffic deaths to historic lows and changed the culture on our streets through a comprehensive program of engineering, education, and enforcement.



Vision Zero convenes agencies from all over the City to work together, break down barriers, and save lives.

Has Vision Zero **improved** the **situation** in NYC since its implementation?

How can we answer this
question?

How can we make it
quantifiable?

Has Vision Zero **improved** the **situation** in NYC since its implementation?



when: Before/after a date or over time?

Where: Everywhere or in some designated regions?

N. of injuries

N. of fatalities

N. of injuries / N. of collisions

N. of fatalities / N. of collisions

N. of speed tickets

Has Vision Zero **improved** the **situation** in NYC since its implementation?



How did the **number of weekly speed tickets** change during the last 2 years in Manhattan?

How did the **monthly number of injuries** change during the last 5 years across the NYC boroughs?

Has Vision Zero **improved** the **situation** in NYC since its implementation?



Not Straight-forward

Is there a **causal relationship** between any of these patterns and the Vision Zero plan?

Getting from the initial question to a causal inference requires rigorous analysis!

Correlation does not imply causation!

Clarity First: Define Your Terms

Smart cities

- What does smart even mean here?
- Smart compared to what baseline?
- Smart for whom and for what purposes?
- Who controls the data and decision-making algorithms?



Clarity First: Define Your Terms

Resilience

- Resilient to what shocks: floods, pandemics, economic downturns, daily disruptions?
- Resilient for whom: commuters, businesses, vulnerable residents, future generations?
- Resilient in what sense: continuity of service, speed of recovery, or capacity to adapt?



Clarity First: Define Your Terms

World-class cities

- World-class by whose standard: global rankings, local residents, or political leaders?
- World-class in what domain: finance, transport, culture, environment?
- World-class compared to what baseline: peer cities regionally, global capitals, or an ideal benchmark?



Problem Definition

- Before starting your data analysis make sure you have:
 - **Clear definitions** of your key terms
 - A clear picture of the **desired outcome**
 - **Measurable objectives** (where, when, for whom?)
- Don't just assume, ask and make clear
- You often end up with several clear questions with quantifiable outcomes instead of one vague question

Tips for your Final Project



In your **final project** submissions:

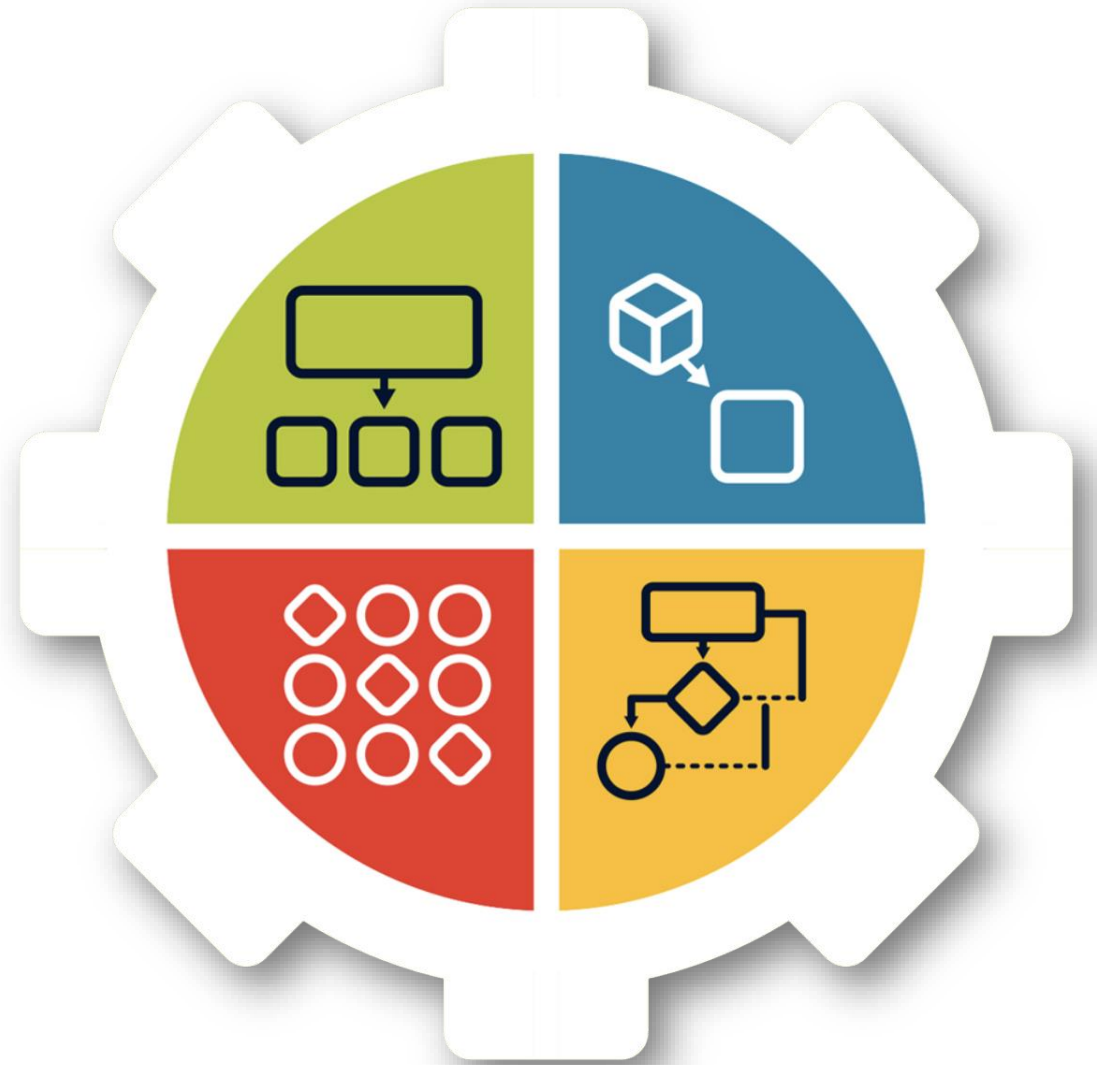
- Define key terms explicitly.
- Cite the source of the definition.
 - Is it from a published paper?
 - From a city agency?
 - From an international standard?
- Explain why you chose it: why does this definition make sense for your project, your context, your audience?

Class Activity

- Form groups of 3-5 members
- On schedule page on the website locate this session
- In resources column, find *Problem Definition Exercise*
- Open Miro Board and follow instructions
- Time: 25 minutes

Computational Thinking Principles

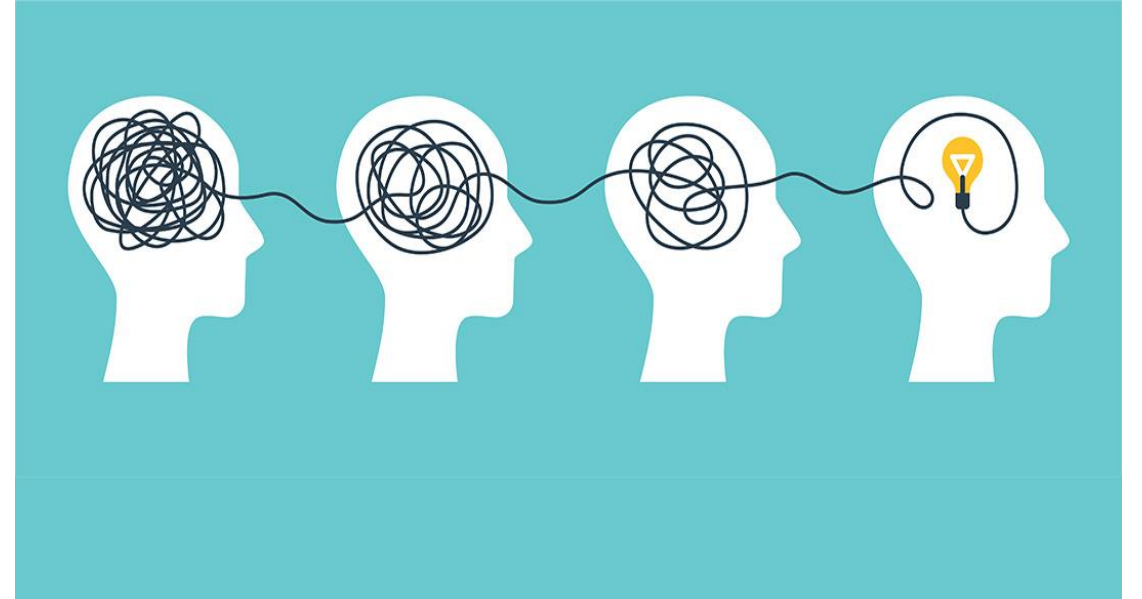
- Decomposition
- Abstraction
- Pattern Recognition
- Algorithm





Decomposition

- Taking a (complex) problem and breaking it into more manageable sub-problems
- Best to break a problem into parts that you already know how to solve
- Creating structures to help us solve the problem

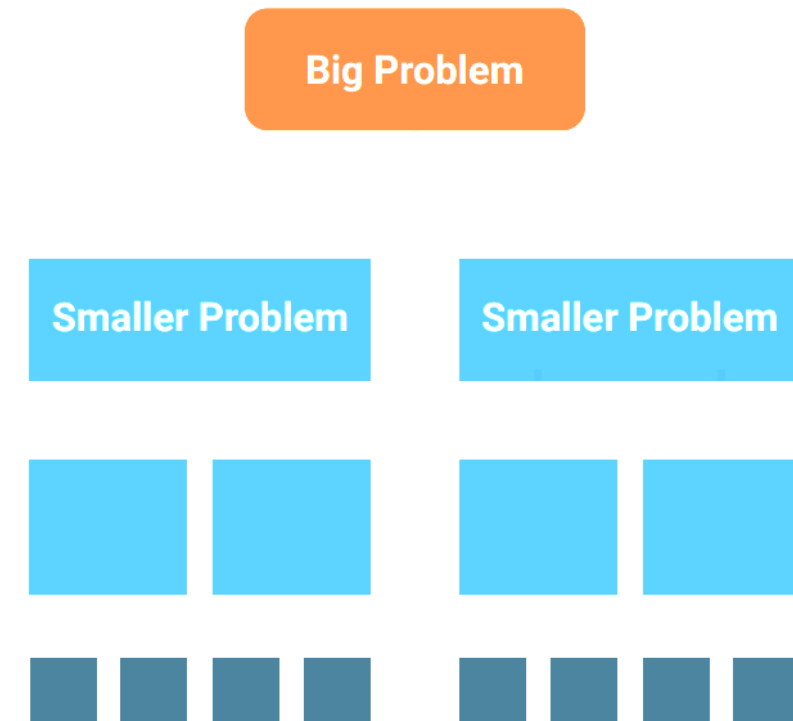




Decomposition

If you can't solve a problem,
then there is an easier problem
you can solve: find it.

GEORGE PÓLYA

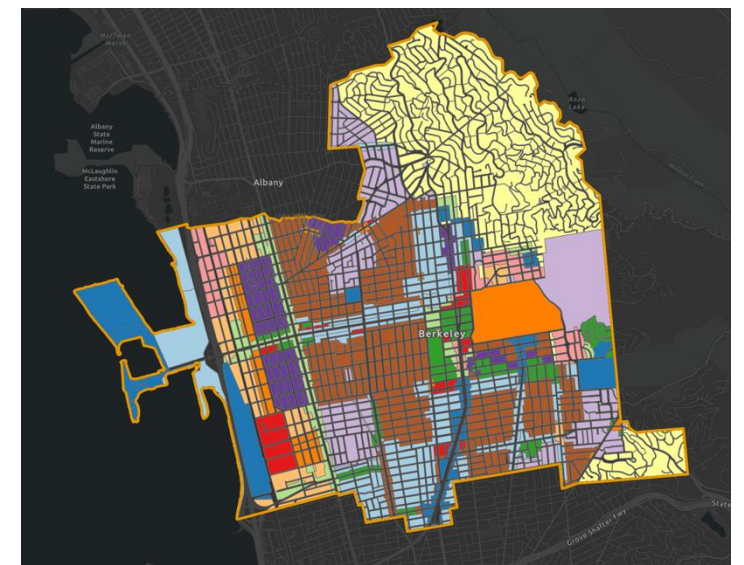




City



Neighborhoods



Zoning Districts

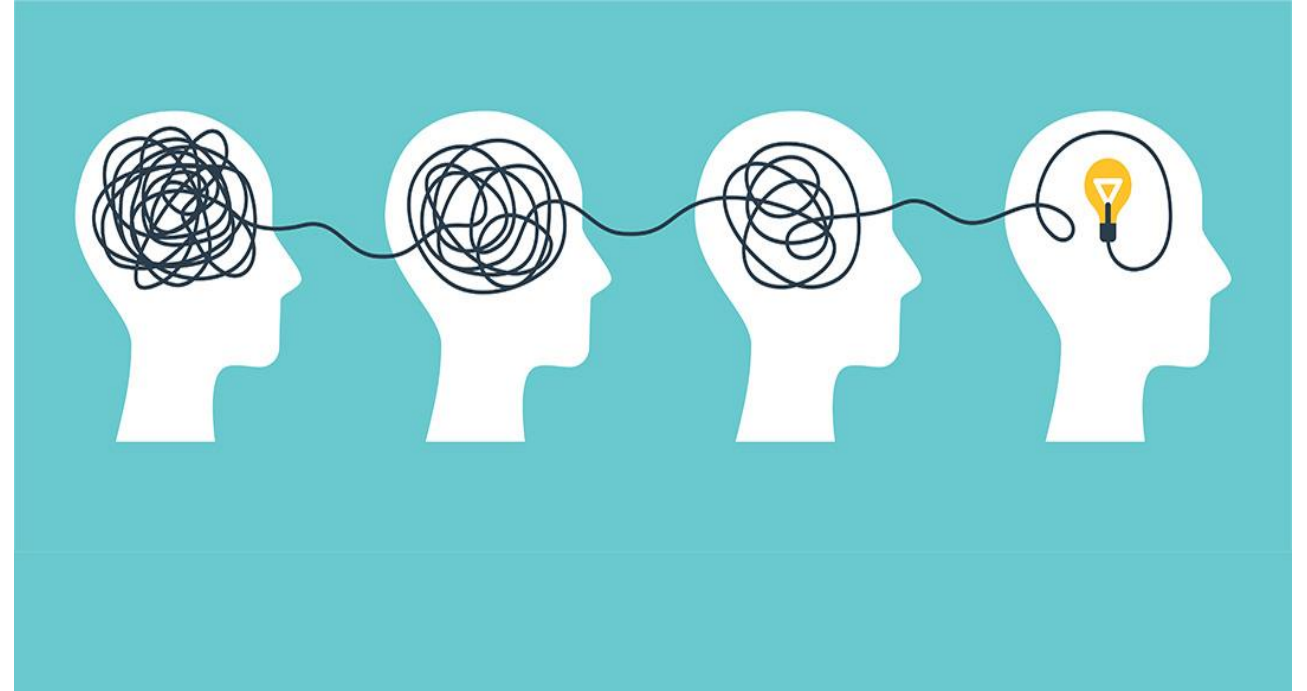
Decomposition

Decompose the same entity in different ways



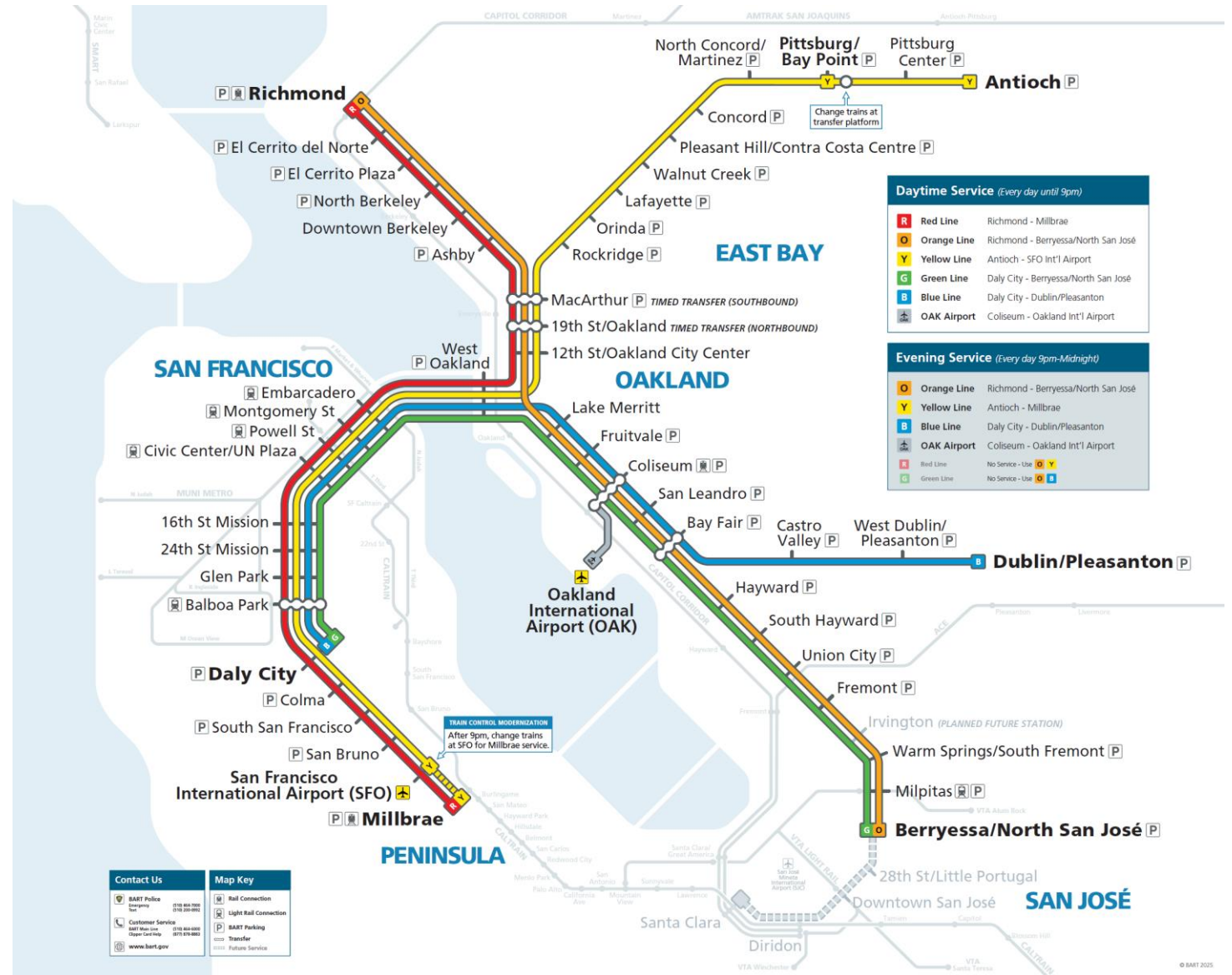
Abstraction

- Representing problems in a simplified, **general** way.
- Focus on the critical aspects of the problem
- Eliminate specific details and irrelevant patterns that do not contribute to solving the problem



Abstraction

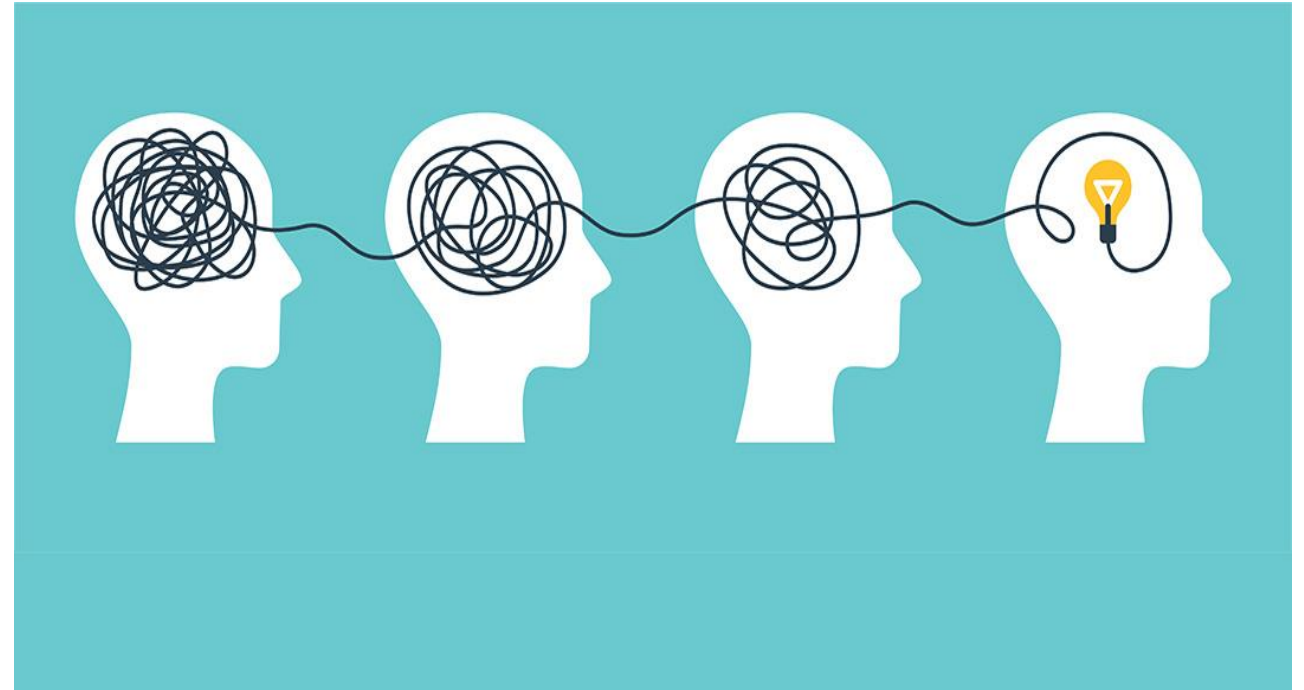
What abstractions were applied to create this map and make it easier to understand?





Pattern Recognition

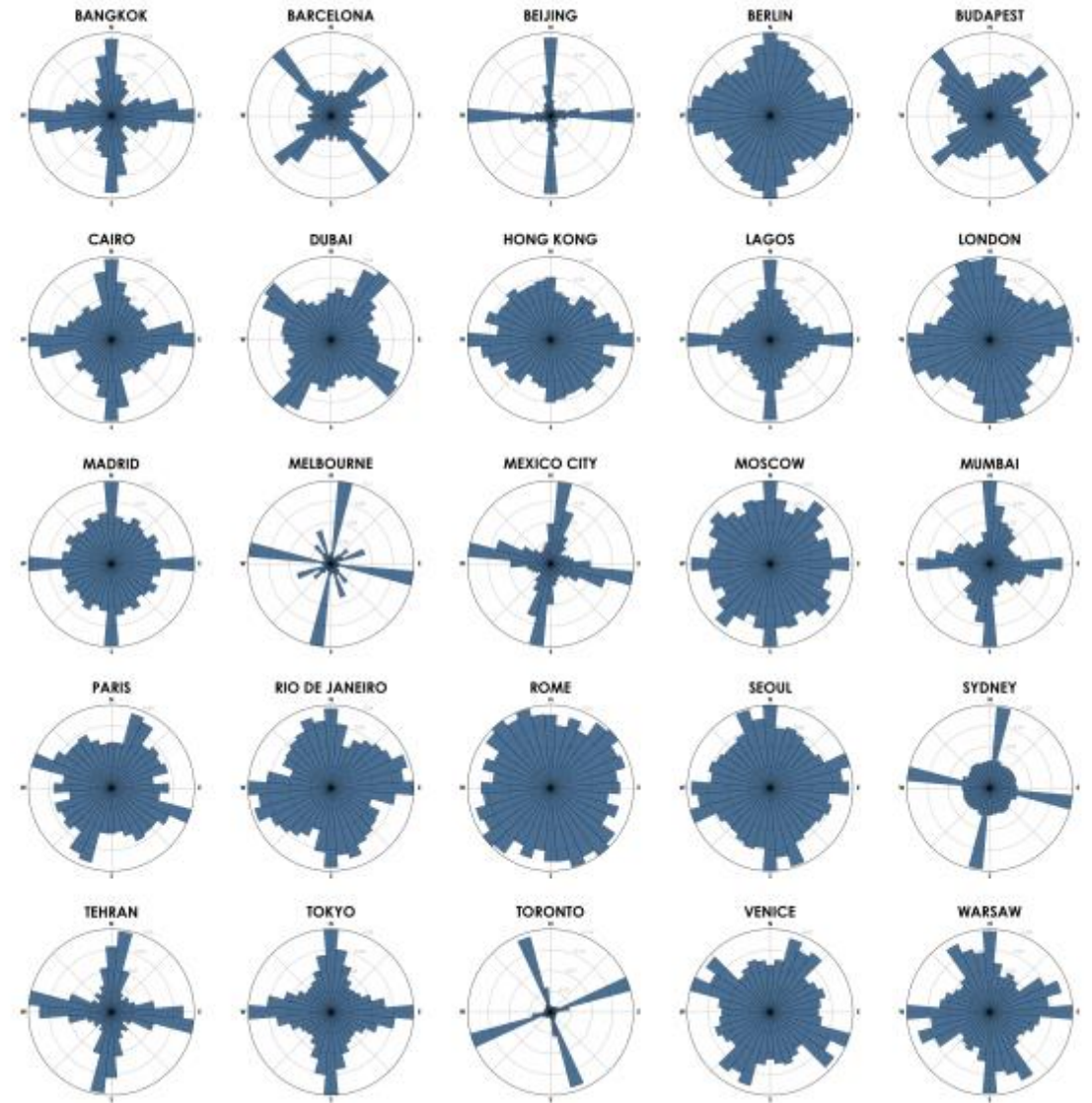
- Identifying patterns or trends within a problem
- Identify specific similarities and differences among similar problems to work towards a solution
- Problems are easier to solve when they share patterns, because we can use the same problem-solving solution wherever the pattern exists.



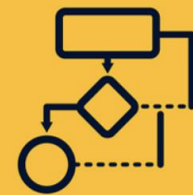
Pattern Recognition

- Can you see common patterns?
- Can we group these cities into distinct categories based on the patterns observed?

City Street Network Orientation

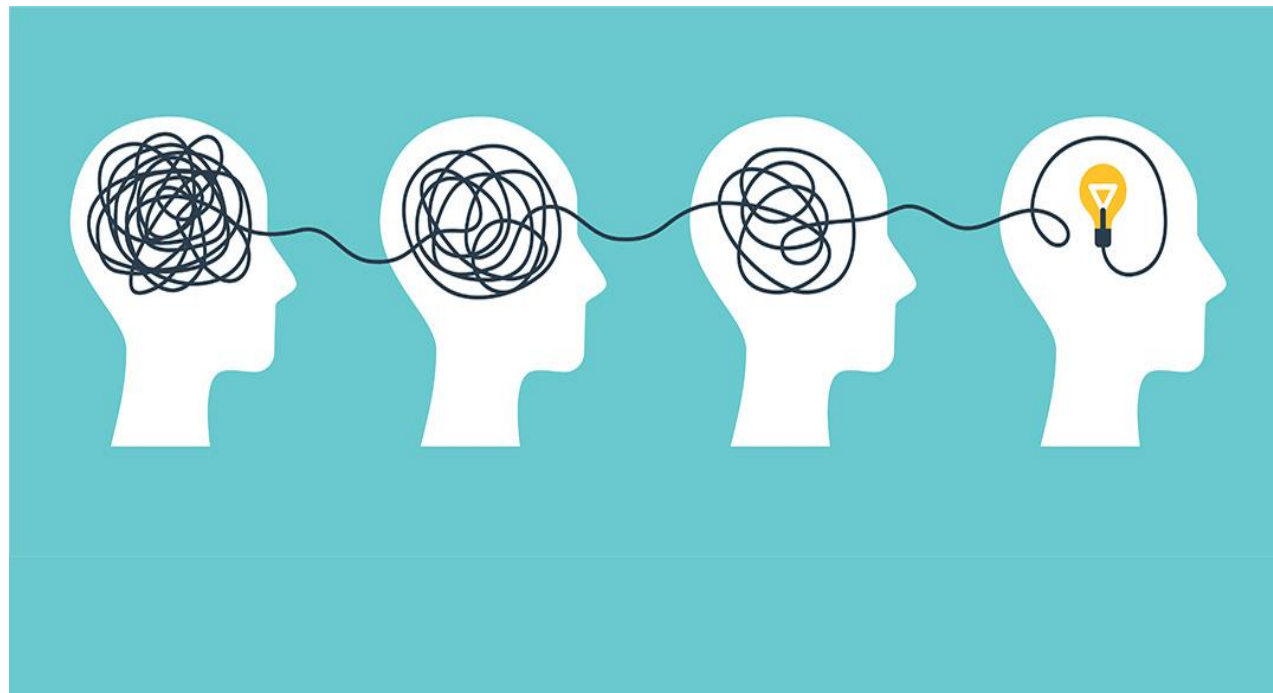


Boeing, 2019



Algorithm

- Step-by-step instructions for solving a problem which can be used for similar problems.
- Following the algorithm's steps correctly ensures arriving at the correct solution, regardless of the input.



Code is an implementation of an algorithm



Turing Complete:

Anything computable in one language, is computable in any other programming languages.